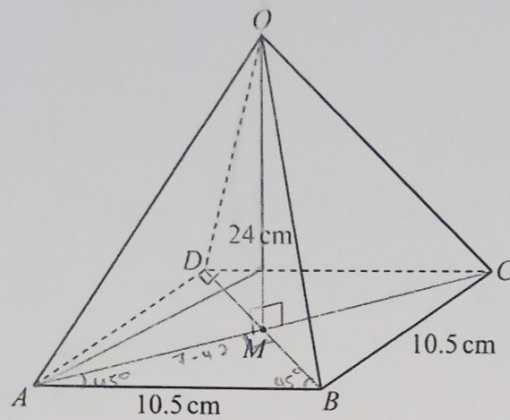




NOT TO
SCALE

The diagram shows a pyramid $OABCD$.
The pyramid has a square base, $ABCD$, with sides 10.5 cm.
The vertex O is vertically above the centre of the base, M .
The height of the pyramid is 24 cm.

(a) Calculate the angle that OA makes with the base.

~~AC~~

OA

$$\sin \theta = \frac{\text{opp}}{\text{hyp}}$$

$$\sin \theta = \frac{\text{opp}}{20.5}$$

$$\sin \theta \times 20.5 = \text{opp}$$

$$7.42 \text{ cm} = \text{opp}$$

$$a^2 + b^2 = c^2$$

$$7.4^2 + 24^2 = c^2$$

$$\sqrt{7.4^2 + 24^2} = c$$

$$25.1 \text{ cm} = OA$$

$$\sin \theta = \frac{\text{opp}}{\text{hyp}}$$

$$\sin \theta = \frac{24}{25.1}$$

$$\theta = \sin^{-1} \left(\frac{24}{25.1} \right)$$

$$\theta = 72.97^\circ$$

$$\dots\dots\dots 72.97^\circ \dots\dots\dots [4]$$

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN